Linear Functions & Graphing

30 problems · A worked example comes before each type. Show every step.

Name: _____ Date: _____

Key Idea

A **linear function** makes a straight line. Two numbers tell you everything about it:

- the **slope** $m = \dfrac{\text{rise}}{\text{run}} = \dfrac{y_2 - y_1}{x_2 - x_1}$ — how steep the line is, and which way it goes;
- the y**-intercept** b — where the line crosses the y-axis, the point $(0, b)$.

Put together: **slope-intercept form** $y = mx + b$.

Common Mistake

Keep the same order on top and bottom: $\dfrac{y_2 - y_1}{x_2 - x_1}$, **not** $\dfrac{y_2 - y_1}{x_1 - x_2}$.

In $y = mx + b$, y must be alone first. In $2y = 4x + 6$ the slope is 2, **not** 4: divide everything by 2 to get $y = 2x + 3$.

Type 1 Slope from Two Points

Watch Me — Worked Example

Find the slope of the line through $(1, 2)$ and $(4, 8)$.

Rise over run: how much y changes, divided by how much x changes.

Step 1 Subtract the y's $8 - 2 = 6$

Step 2 Subtract the x's (same order) $4 - 1 = 3$

Answer $\boldsymbol{m = \dfrac{6}{3} = 2}$

1. Find the slope through $(2, 3)$ and $(4, 7)$.

$m =$ _____

2. Find the slope through $(1, 5)$ and $(3, 1)$.

$m =$ _____

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3. Find the slope through $(-2, 4)$ and $(2, 6)$.

$m =$ _____

4. Find the slope through $(0, -3)$ and $(5, -3)$.

$m =$ _____

5. Find the slope through $(-4, -1)$ and $(2, 8)$.

$m =$ _____

6. Find the slope through $(3, 2)$ and $(3, 7)$.

$m =$ _____

Type 2 Slope from a Graph

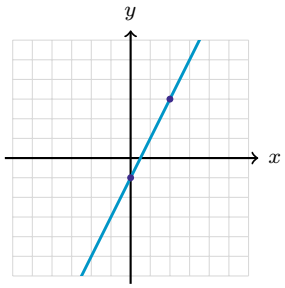
Watch Me — Worked Example

Find the slope of the line in the graph.

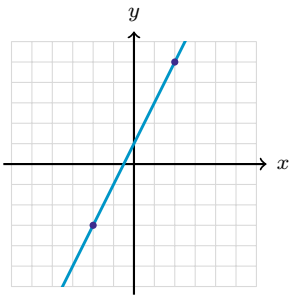
Pick two points where the line crosses grid corners: $(0, -1)$ and $(2, 3)$. Count from the left point to the right point.

Up 4 (rise), right 2 (run).

Answer $m = \dfrac{4}{2} = 2$. Going up to the right means a **positive** slope.

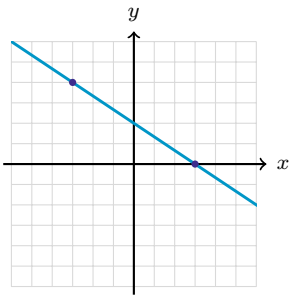


7. Find the slope of the line.



$m =$ _____

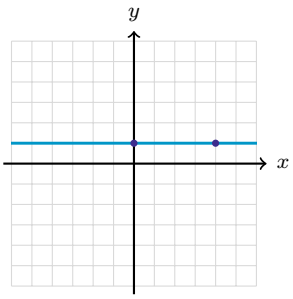
8. Find the slope of the line.



$m =$ _____

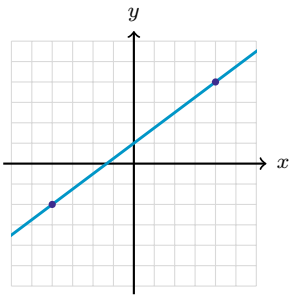
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9. Find the slope of the line.



$m =$ _____

10. Find the slope of the line.



$m =$ _____

Type 3 Intercepts

Watch Me — Worked Example

Find the x- and y-intercepts of $2x + 3y = 12$.

On the x-axis, $y = 0$. On the y-axis, $x = 0$.

x-intercept **Let** $y = 0$ $\quad 2x = 12,\ x = 6 \rightarrow (6, 0)$

y-intercept **Let** $x = 0$ $\quad 3y = 12,\ y = 4 \rightarrow (0, 4)$

Answer $\quad \mathbf{(6, 0)}$ and $\mathbf{(0, 4)}$

11. Find both intercepts of $2x + 3y = 12$.

x-int, y-int: _____

12. Find both intercepts of $3x - 4y = 12$.

x-int, y-int: _____

13. Find both intercepts of $5x + 2y = 10$.

x-int, y-int: _____

14. Find both intercepts of $-4x + 6y = 24$.

x-int, y-int: _____

15. Find both intercepts of $x + 2y = -8$.

x-int, y-int: _____

16. Find both intercepts of $3x + 5y = 15$.

x-int, y-int: _____

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Type 4 Slope-Intercept Form $y = mx + b$

Watch Me — Worked Example

Write $4x + 2y = 6$ in slope-intercept form. Name m and b.

Get y alone, exactly like a literal equation.

Step 1 Subtract $4x$ $\quad 2y = -4x + 6$

Step 2 Divide *every* term by 2 $\quad y = -2x + 3$

Answer $\quad \boldsymbol{y = -2x + 3,\ m = -2,\ b = 3}$

17. $y = 3x - 4$
Write in $y = mx + b$ form, then name m and b.

m, b: _____

18. $y = -\frac{1}{2}x + 5$
Write in $y = mx + b$ form, then name m and b.

m, b: _____

19. $2x + y = 7$
Write in $y = mx + b$ form, then name m and b.

m, b: _____

20. $3x + 3y = 9$
Write in $y = mx + b$ form, then name m and b.

m, b: _____

21. $4x - 2y = 8$
Write in $y = mx + b$ form, then name m and b.

m, b: _____

22. $x - 3y = 6$
Write in $y = mx + b$ form, then name m and b.

m, b: _____

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Type 5 Graph $y = mx + b$

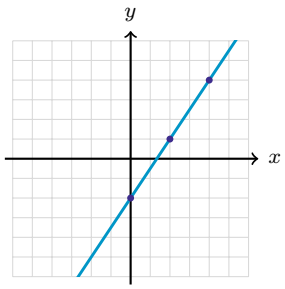
Watch Me — Worked Example

Graph $y = \frac{3}{2}x - 2$.

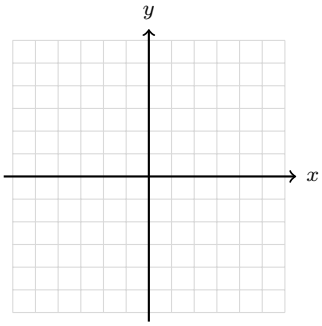
Step 1 Start at b: plot $(0, -2)$.

Step 2 Slope $\frac{3}{2}$: up 3, right 2 $\rightarrow (2, 1)$.

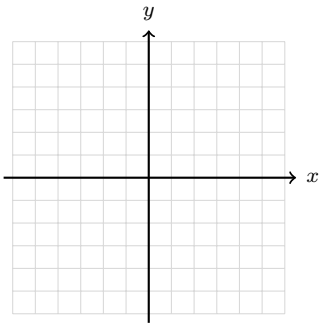
Step 3 Do it again $\rightarrow (4, 4)$, then draw the line.



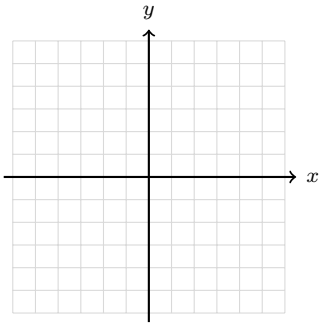
23. Graph $y = 2x - 3$.



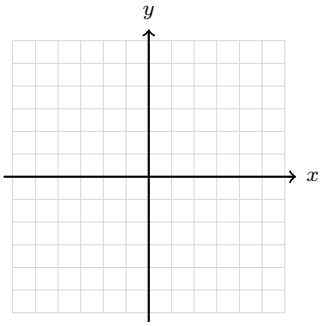
24. Graph $y = -\frac{1}{2}x + 4$.



25. Graph $y = \frac{2}{3}x - 1$.



26. Graph $y = -3x + 2$.



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Type 6 Linear Functions in Real Life

Watch Me — Worked Example

A gym charges a \$20 sign-up fee plus \$5 per visit. Write a function for the cost C after x visits. What do 8 visits cost?

The **starting amount** is b. The amount added **each time** is the slope m.

Step 1 Starting amount $b = 20$

Step 2 Rate (per visit) $m = 5$

Step 3 Write and use it $C = 5x + 20;\ C = 5 \cdot 8 + 20 = 60$

Answer $\boldsymbol{C = 5x + 20}$; 8 visits cost $\boldsymbol{\$60}$

27. A taxi charges \$3 to start plus \$2 per mile. Write the cost C for x miles. Find the cost of a 12-mile ride.

Answer: _____

28. A candle is 30 cm tall and burns 2 cm per hour. Write its height h after t hours. When does it burn out?

Answer: _____

29. A plant is 4 inches tall and grows 1.5 inches per week. Write its height h after w weeks. How tall is it after 6 weeks?

Answer: _____

30. Mia has \$200 and spends \$15 each week. Write her money M after w weeks. What do the slope and y-intercept mean?

Answer: _____

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